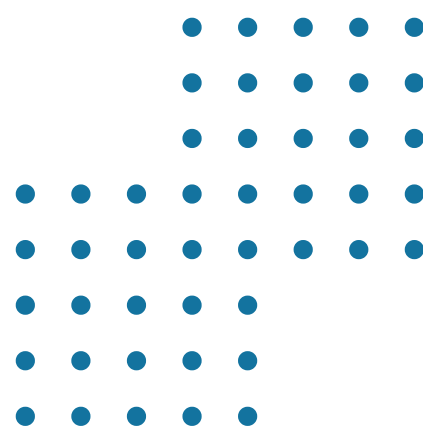
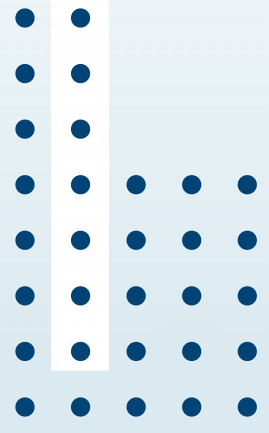




LOCAL BUSINESS SUPPORT RECOVERY

Team Members:

- Sara Sofia Socha Gil
 - Daniel Eduardo Rincón Vivas
 - Diego Alejandro Yañes Zabala
 - Sebastian Zambrano Hurtado
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Today's Agenda

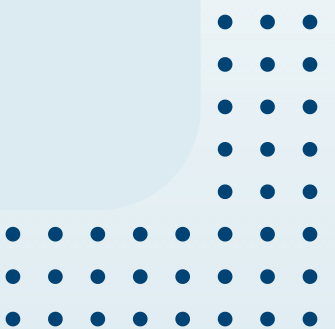
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|-----------|---|-----------|--------------------------|
| 01 | The Problem | 06 | System Analysis Findings |
| 02 | Context & Research Site | 07 | Proposed Solution |
| 03 | Methodology | 08 | Key Features |
| 04 | Key Data:
Foot Traffic & Visit Motives | 09 | Project Lifecycle |
| 05 | Key Data:
Spending & Inventory | 10 | Simulation Results |



The Problem

Small Businesses Are Struggling Post-Pandemic

- 41.18% of mall visitors spend less than before the pandemic
- Foot traffic is unpredictable: Sunday brings 2x more visitors than Saturday
- Small stores operate as isolated agents - no real-time data
- Large chains (Falabella, Éxito) dominate through digital tools; small shops rely on foot traffic alone
- 80% of small stores report fast/very fast inventory rotation → stockout risk is high



Context & Research Site

Why Titan Plaza?



- Located on Av. Boyacá con Calle 80, Bogotá — one of the most visited malls in the city
- 4 floors, 250 stores, daily 8am–10pm
- Attracts middle-income population (strata 3–5)
- Contains a mix of large chains + small independent businesses
- Ideal case study: concentrated, observable, real-world environment



Methodology

How We Analyzed the System

Direct Observation

Saturday 9am–12pm & Sunday
12pm–3pm

Counted foot traffic at small store
areas

Surveys

17 customers + 5 employees

Spending habits, visit motives,
inventory

Environmental Correlation

Weather, day of week, promotions

Two observation sessions

Key Data: Foot Traffic & Visit Motives

What the Data Tells Us

Day	Enter	Exit
Saturday (9am - 12pm)	343	262
Sunday (12pm - 3pm)	811	309
Total	1154	571

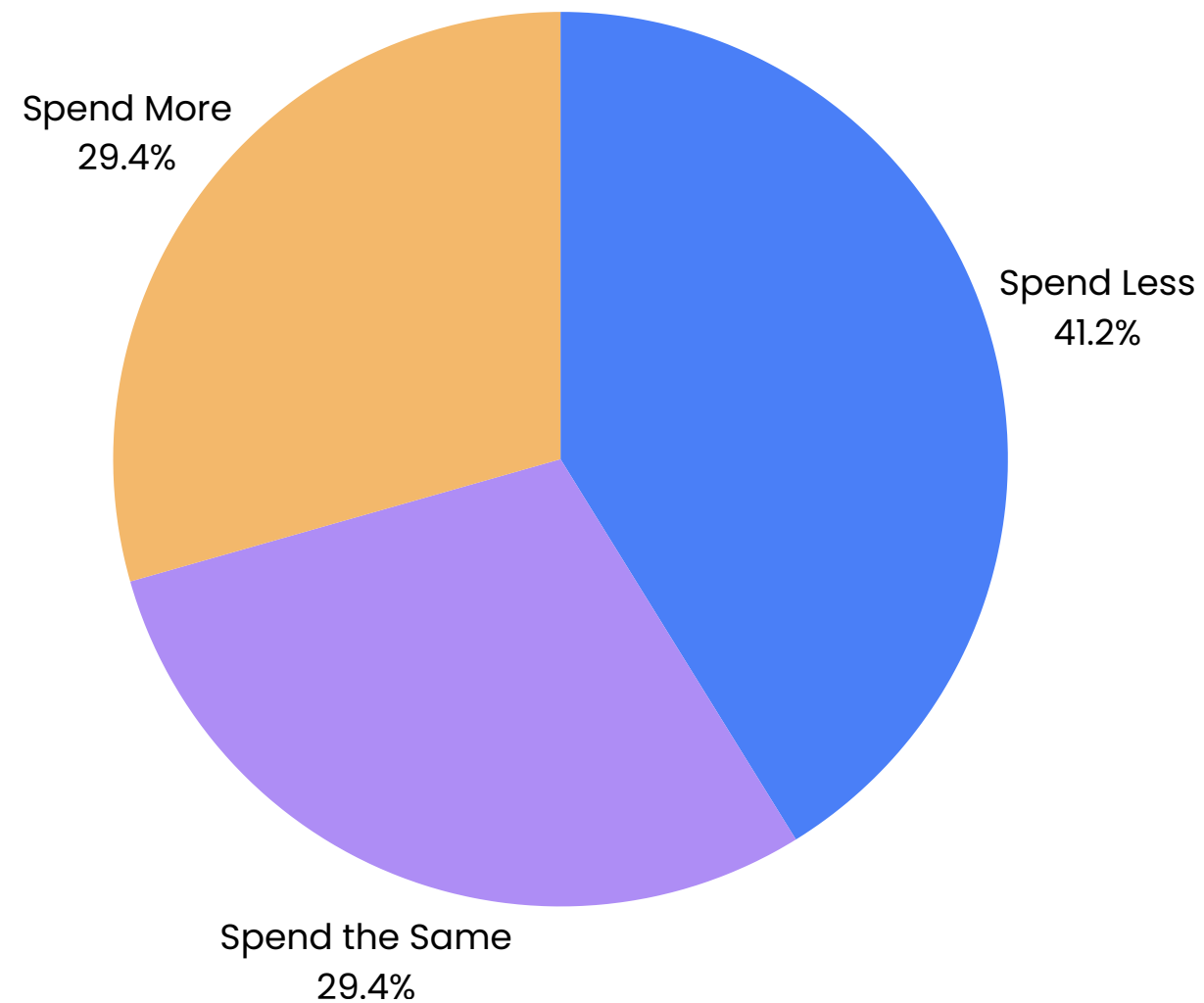
Reasons for visit (top results)

- 29.41% – Other reasons
- 17.65% – Food court / Cinema
- 11.76% – Small / local stores only

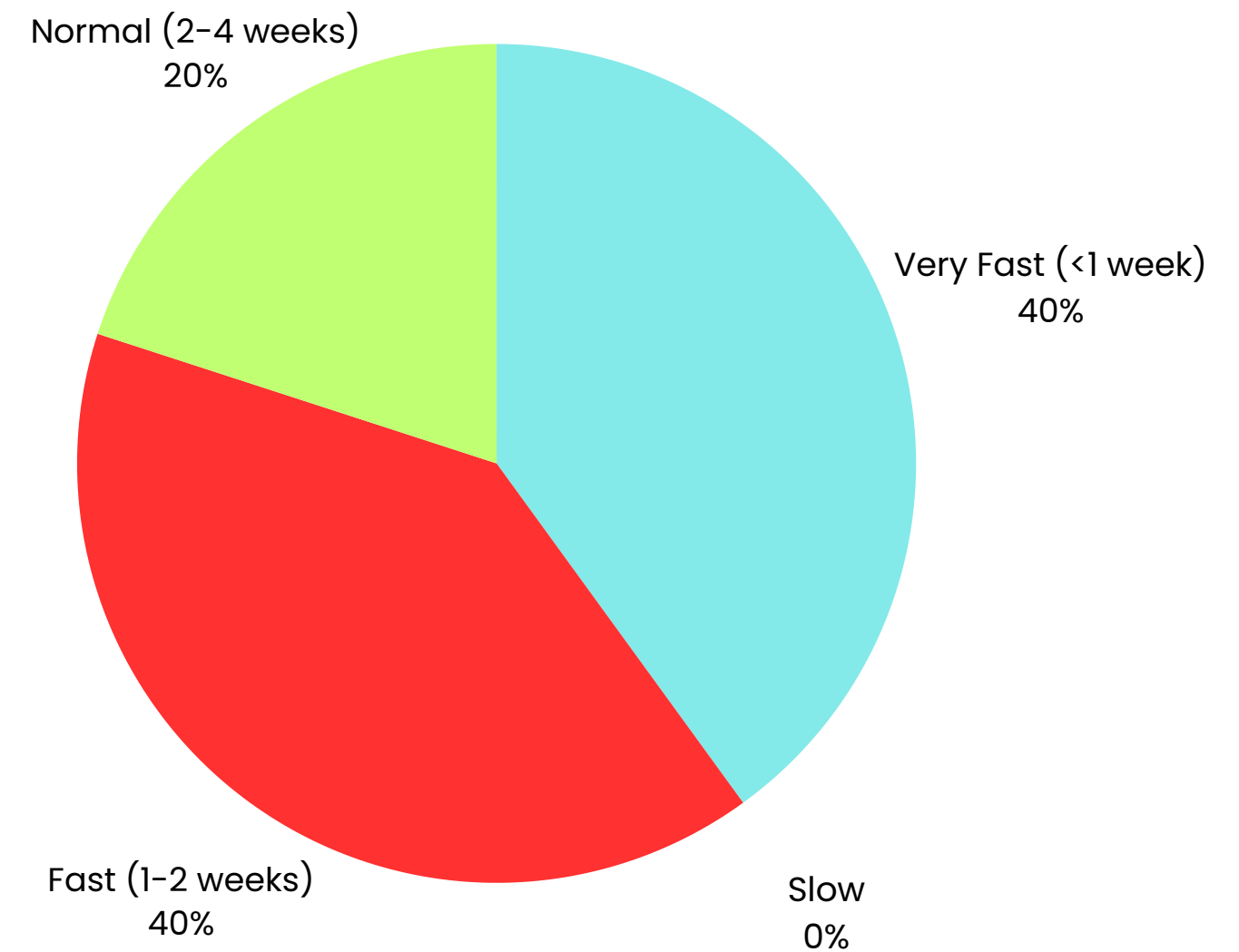
Key Data: Spending & Inventory

The Core Challenge in Numbers

Post-Pandemic Spending:



Inventory Rotation:



System Analysis Findings

What's Really Broken?

Visibility Gap

Small stores have no digital presence customers go straight to large chains or e-commerce.



Feedback Loop Problem

One bad experience (item out of stock) permanently redirects customers to anchor stores.



Informational Disconnect

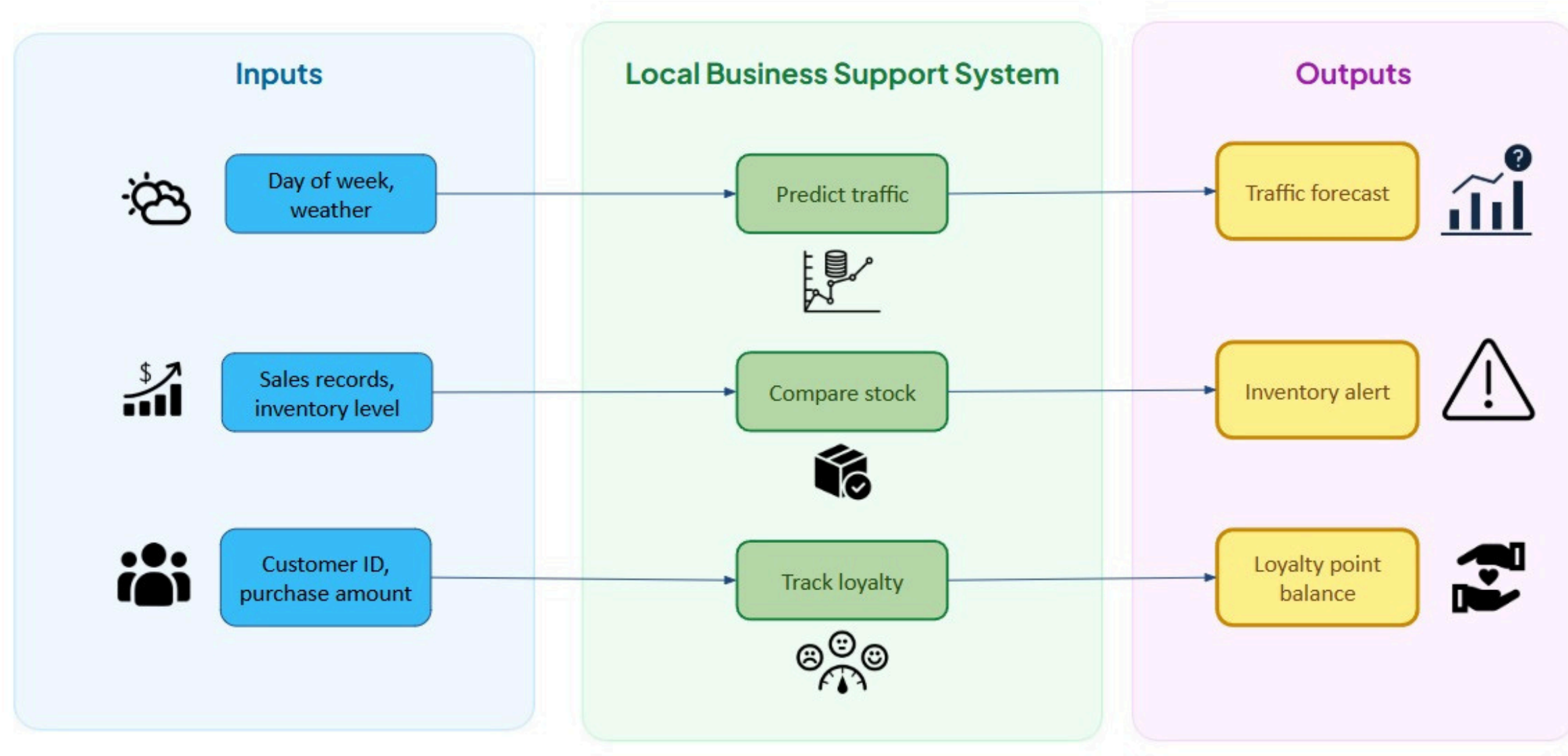
Each shop operates in isolation. No shared data on traffic patterns, demand trends, or customer needs.

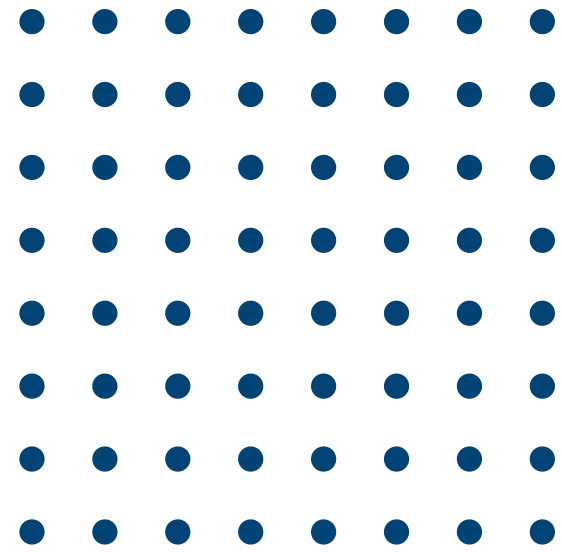


Proposed Solution

Initial System Design

A system that connects customer traffic, inventory levels, and environmental conditions to help small business owners make informed decisions using real data.

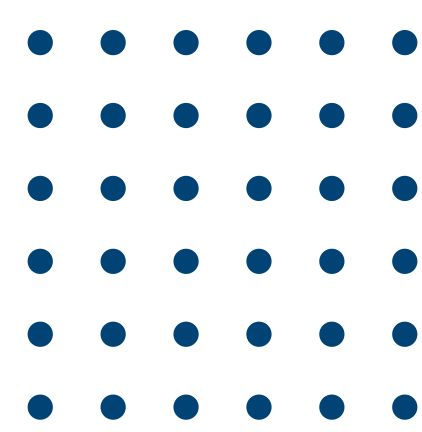




Key Features

What the Platform Does

- **Owner Dashboard:** Real-time metrics – sales, traffic, inventory turnover
- **Inventory Alerts:** Notifies owners before stock runs out
- **Automated Promotions:** Triggered by traffic peaks and low-sales periods
- **Traffic Prediction:** Based on day of week, weather, and historical patterns
- **Product Recommendations:** Personalized suggestions based on purchase trends



Project Lifecycle Phases

Phase 1

Problem
Analysis and
Initial System
Design

Phase 2

Baseline
Simulation
Current System

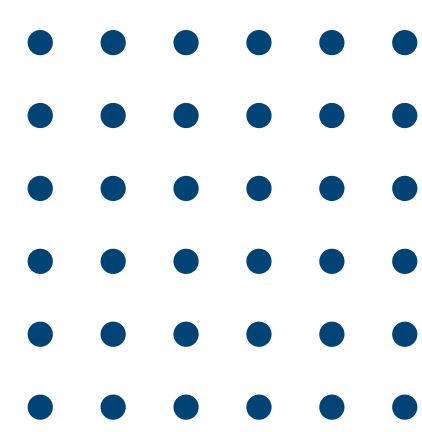
Phase 3

Process
Simulation
Proposed
System

Phase 4

Optimization
Simulation –
Future
Optimized
System.





Simulation Results

baseline scenario maintaining
stability.

(In 100 days)

Optimization

1200 Sales

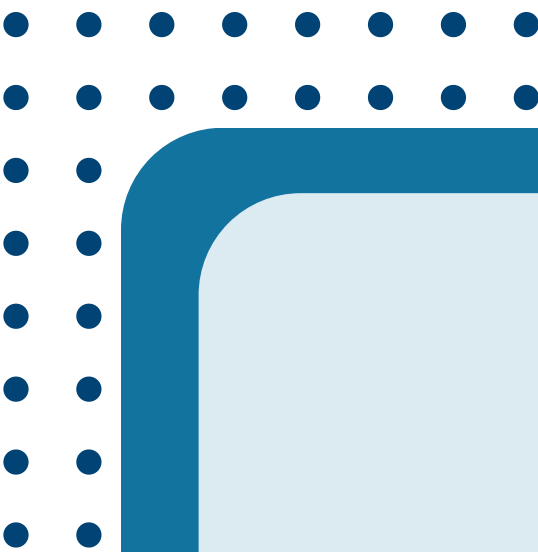
Base Line

510 Sales

Failure

420 Sales

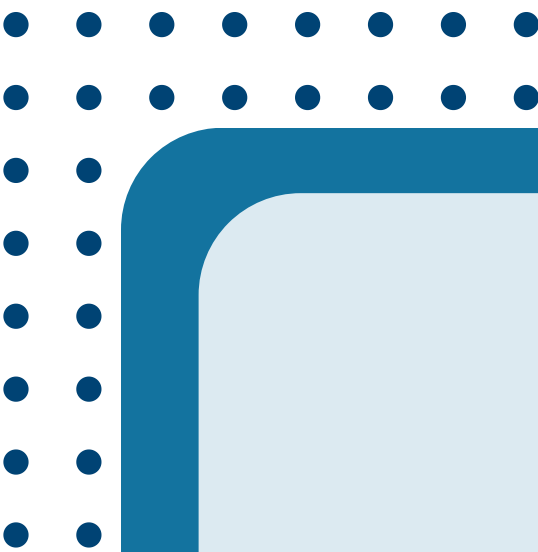




Conclusions

The main challenge faced by small businesses is not the lack of customers, but the lack of information needed to convert foot traffic into sales. The proposed platform integrates customer, inventory, and sales data to support decision-making, and the simulations validated that a data-driven approach can improve sales performance, reduce stockout risks, and increase operational efficiency

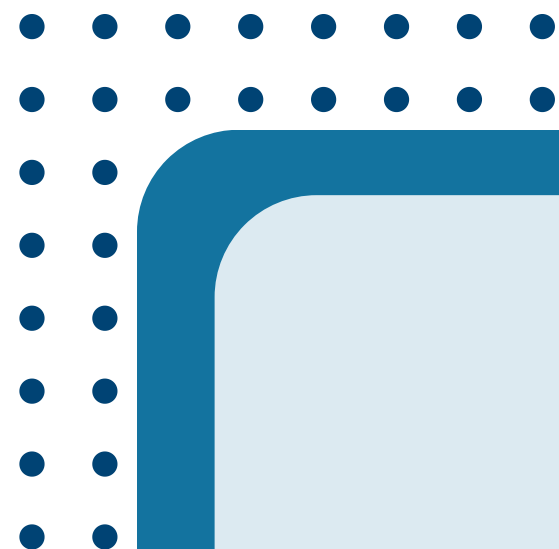




Future Work

we could focus on deploying and validating the platform in a real-world environment, incorporating loyalty programs and customer-facing features, improving demand forecasting through advanced analytics and machine learning, and expanding the solution to other shopping malls to enable cross-mall benchmarking and broader business insights.





THANK YOU

